

Please check the examination details below before entering your candidate information

Candidate surname

Other names

Centre Number

Candidate Number

Pearson Edexcel Award

Tuesday 27 June 2023

Afternoon (Time: 3 hours)

Paper reference

9811/01

Advanced Extension Award

Mathematics

You must have:

Mathematical Formulae and Statistical Tables

An insert for questions 5, 6 and 7 (enclosed)

Total Marks

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions.
- Answer the questions in the spaces provided – *there may be more space than you need.*
- **Calculators may not be used.**
- You must **show all your working.**
- Answers should be given in as simple a form as possible, e.g. $\frac{2\pi}{3}$, $\sqrt{2}$, $3\sqrt{2}$.

Information

- The total mark for this paper is 100 of which **7** marks are for style and clarity of presentation.
- The style and clarity of presentation marks will be indicated as **(+S1) or (+S2)**.
- There are 7 questions in this question paper.
- The marks for each question are shown in brackets.
- The total mark for each question is shown at the end of the question.

Advice

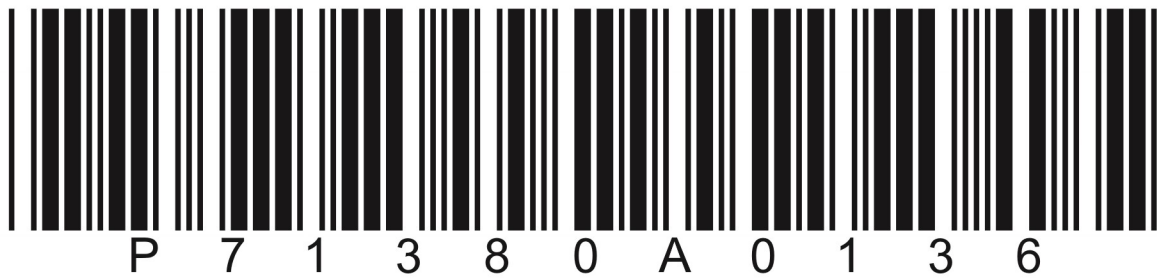
- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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1. (a) Write down the exact value of $\cos 405^\circ$

(1)

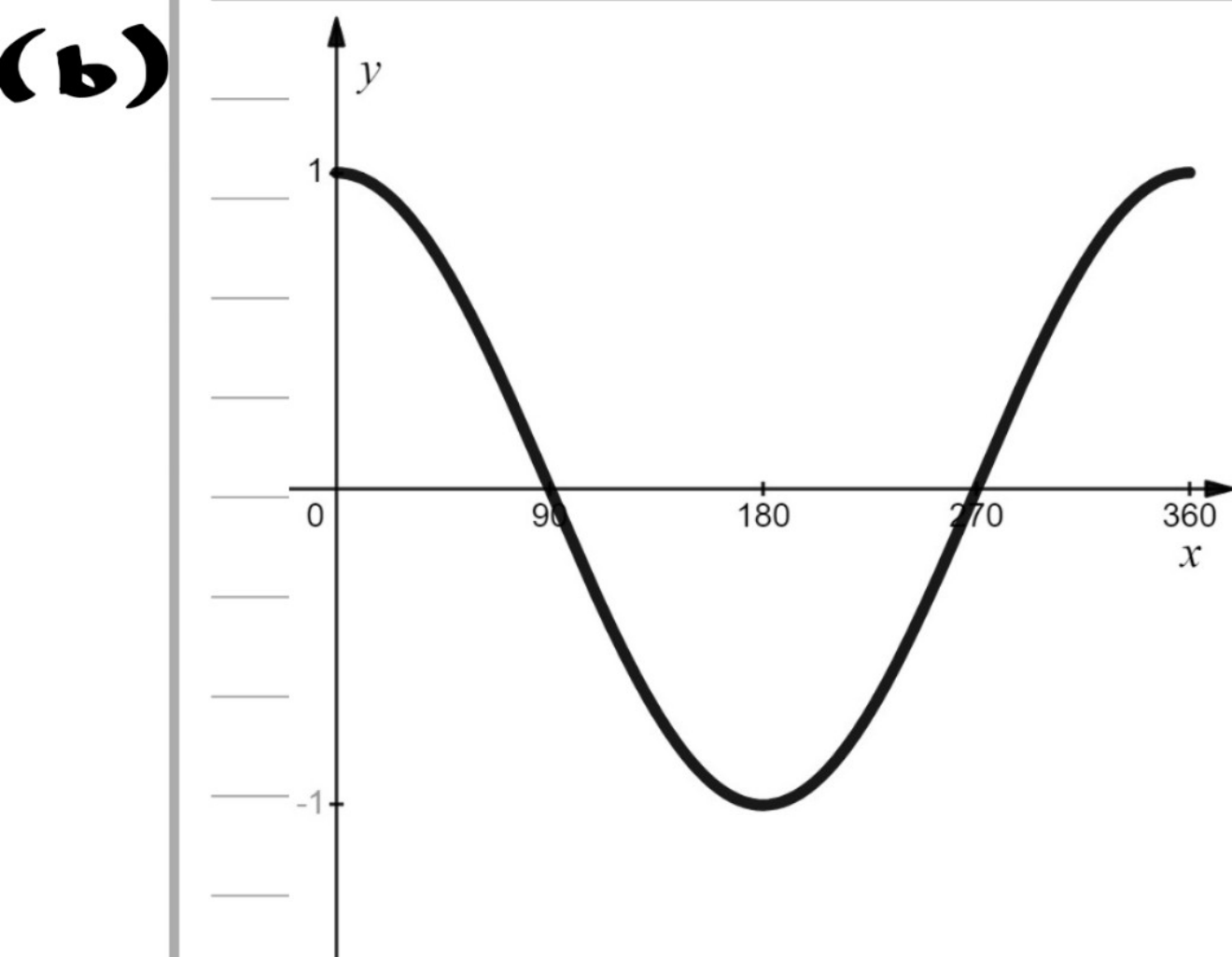
(b) Hence, using a double angle identity for cosine, or otherwise, determine the exact value of $\cos 101.25^\circ$, giving your answer in the form

$$a\sqrt{b + c\sqrt{2 + \sqrt{2}}}$$

where a , b and c are rational numbers.

(5)

(a) $\cos 405^\circ = \cos (45^\circ + 360^\circ)$
 $= \frac{\sqrt{2}}{2}$



NOTE:

$$\cos^2 x = \frac{1}{2}(1 + \cos 2x)$$

$$\therefore \cos x = \pm \sqrt{\frac{1}{2}(1 + \cos 2x)}$$

$$\cos 202.5^\circ = -\sqrt{\frac{1}{2}(1 + \cos 405^\circ)}$$

$$= -\sqrt{\frac{1}{2}\left\{1 + \frac{\sqrt{2}}{2}\right\}}$$

$$= -\sqrt{\frac{1}{4}(2 + \sqrt{2})}$$

$$= -\frac{1}{2}\sqrt{2 + \sqrt{2}}$$

$$\cos 101.25^\circ = -\sqrt{\frac{1}{2}(1 + \cos 202.5^\circ)}$$

Question 1 continued

$$= -\sqrt{\frac{1}{2}\left(1 - \frac{1}{2}\sqrt{2 + \sqrt{2}}\right)}$$

$$= -\sqrt{\frac{1}{4}\left(2 - \sqrt{2 + \sqrt{2}}\right)}$$

$$= -\frac{1}{2}\sqrt{2 - \sqrt{2 + \sqrt{2}}}$$

$$\therefore a = -\frac{1}{2}, b = 2, c = -1$$

(Total for Question 1 is 6 marks)



P 7 1 3 8 0 A 0 3 3 6

2. A student is attempting to prove that there are infinitely many prime numbers.

The student's attempt to prove this is in the box below.

Assume there are only finitely many prime numbers, then there is a biggest prime number, p .

Let $n = 2p + 1$. Then n is bigger than p and since $2p + 1$ is not divisible by p , n is a prime number.

Hence n is a prime number bigger than p , contradicting the initial assumption. So we conclude there are infinitely many prime numbers.

- (a) Use $p = 7$ to show that the following claim made in the student's proof is **not** true:

since $2p + 1$ is not divisible by p , n is a prime number.

(1)

The student changes their proof to use $n = 6p + 1$ instead of $n = 2p + 1$

- (b) Show, by counter example, that this does not correct the student's proof.

(2)

- (c) Write out a correct proof by contradiction to show that there are infinitely many prime numbers.

(5)

(+S1)

(a) $p = 7: n = 2 \times 7 + 1 = 15$

ALTHOUGH 15 IS NOT DIVISIBLE BY 7,

$15 = 5 \times 3$ SO NOT PRIME

$\therefore$ CLAIM IS NOT TRUE

(b) $p = 19: n = 6 \times 19 + 1 = 115 = 5 \times 23$

$\therefore$ THIS DOES NOT CORRECT PROOF

(c) ASSUME THERE ARE FINITELY MANY PRIMES,

THEN WE CAN WRITE A COMPLETE LIST OF PRIMES

LET $P_1, P_2, \dots, P_n$ BE A COMPLETE LIST OF PRIMES



Question 2 continued

CONSIDER THE NUMBER: $N = p_1 \times p_2 \times \dots \times p_n + 1$

NOTE: AS 1 IS NOT PRIME, ADDING 1 GIVES A REMAINDER OF 1, WHICH ENSURES NONE OF THESE PRIME FACTORS $p_1, p_2, \dots, p_n$ DIVIDE INTO THIS NEW NUMBER N

EITHER N IS PRIME OR N IS NOT PRIME

IF N IS PRIME, THEN N IS NOT IN THE LIST

$\therefore$ THIS IS A CONTRADICTION $\therefore p_1, p_2, \dots, p_n$ IS A COMPLETE LIST OF PRIMES

IF N IS NOT PRIME, THROUGH THE FACTOR DECOMPOSITION, N HAS A PRIME FACTOR WHICH MUST BE A FACTOR OF 1

THIS IS IMPOSSIBLE $\therefore$ THIS IS A CONTRADICTION

$\therefore$ THERE ARE INFINITELY MANY PRIMES



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3.

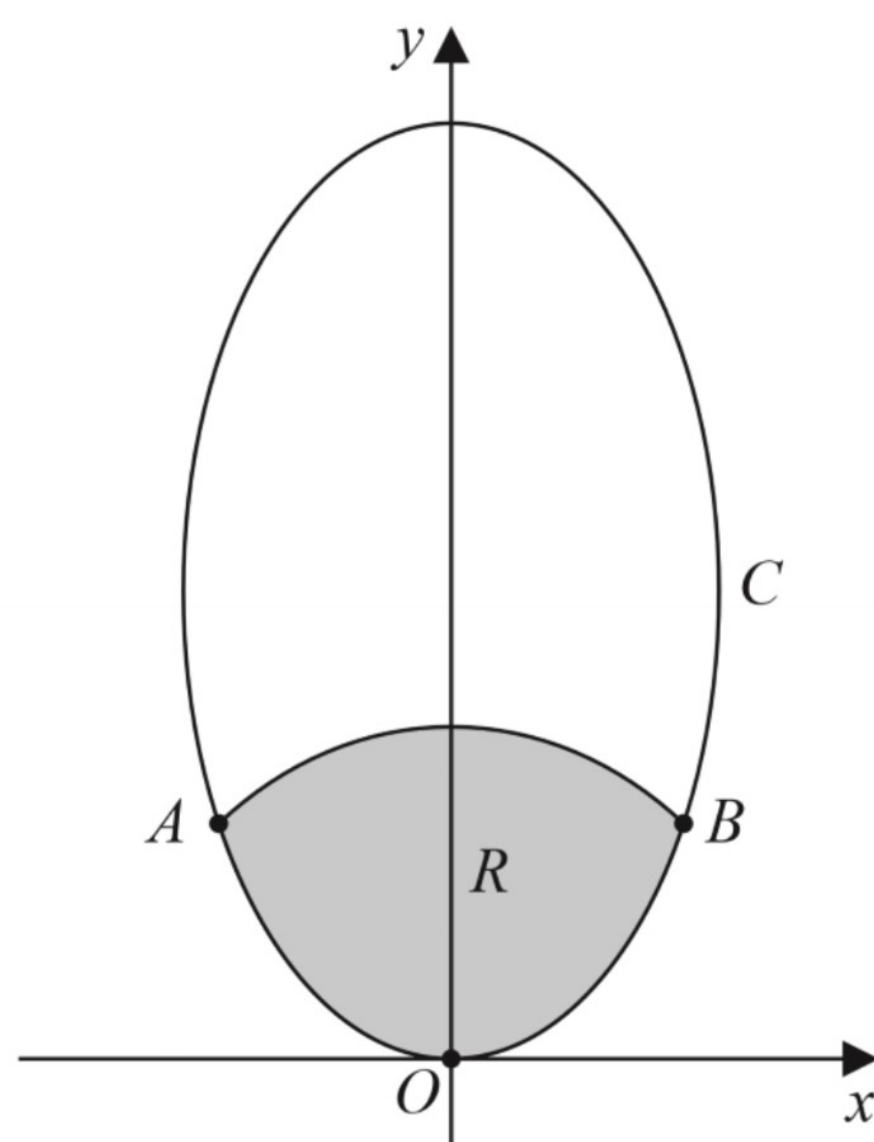


Figure 1

Figure 1 shows the curve C given by the parametric equations

$$x = \frac{5}{\sqrt{3}} \sin t \quad y = 5(1 - \cos t) \quad 0 \leq t \leq 2\pi$$

The circle with centre at the origin O and with radius $\frac{5\sqrt{2}}{2}$ meets the curve C at the points A and B as shown in Figure 1.

(a) Determine the value of t at the point B .

(3)

The region R , shown shaded in Figure 1, is bounded by the curve C and the circle.

(b) Determine the area of the region R .

(7)

(a) $x^2 + y^2 = \left(\frac{5\sqrt{2}}{2}\right)^2$

$$x^2 + y^2 = \frac{25}{2}$$

$$\left\{\frac{5}{\sqrt{3}} \sin t\right\}^2 + \{5(1 - \cos t)\}^2 = \frac{25}{2}$$

$$\frac{25}{3} \sin^2 t + 25(1 - 2\cos t + \cos^2 t) = \frac{25}{2} \quad \times \frac{6}{25}$$



Question 3 continued

$$2\sin^2 t + 6 - 12\cos t + 6\cos^2 t = 3$$

$$2(1 - \cos^2 t) + 6 - 12\cos t + 6\cos^2 t = 3$$

$$4\cos^2 t - 12\cos t + 5 = 0$$

$$(2\cos t - 1)(2\cos t - 5) = 0$$

$$\cos t = \frac{1}{2}, \cos t \neq \frac{5}{2}$$

AT B, x IS POSITIVE

$$\therefore t = \frac{\pi}{3}$$

(b) AREA IS GIVEN BY $\int_a^b y \frac{dx}{dt} dt$

$$\frac{dx}{dt} = \frac{5}{\sqrt{3}} \cos t$$

$$\text{AREA UNDER C} = 2 \int_0^{\frac{\pi}{3}} 5(1 - \cos t) \frac{5}{\sqrt{3}} \cos t dt$$

$$= \frac{50}{\sqrt{3}} \int_0^{\frac{\pi}{3}} [\cos t - \cos^2 t] dt$$

$$= \frac{50}{\sqrt{3}} \int_0^{\frac{\pi}{3}} \left[\cos t - \frac{1}{2}(1 + \cos 2t) \right] dt$$

$$= \frac{50}{\sqrt{3}} \left[\sin t - \frac{t}{2} - \frac{\sin 2t}{4} \right]_0^{\frac{\pi}{3}}$$

$$= \frac{50}{\sqrt{3}} \left[\frac{\sqrt{3}}{2} - \frac{\pi}{6} - \frac{\sqrt{3}}{8} \right]$$

Question 3 continued

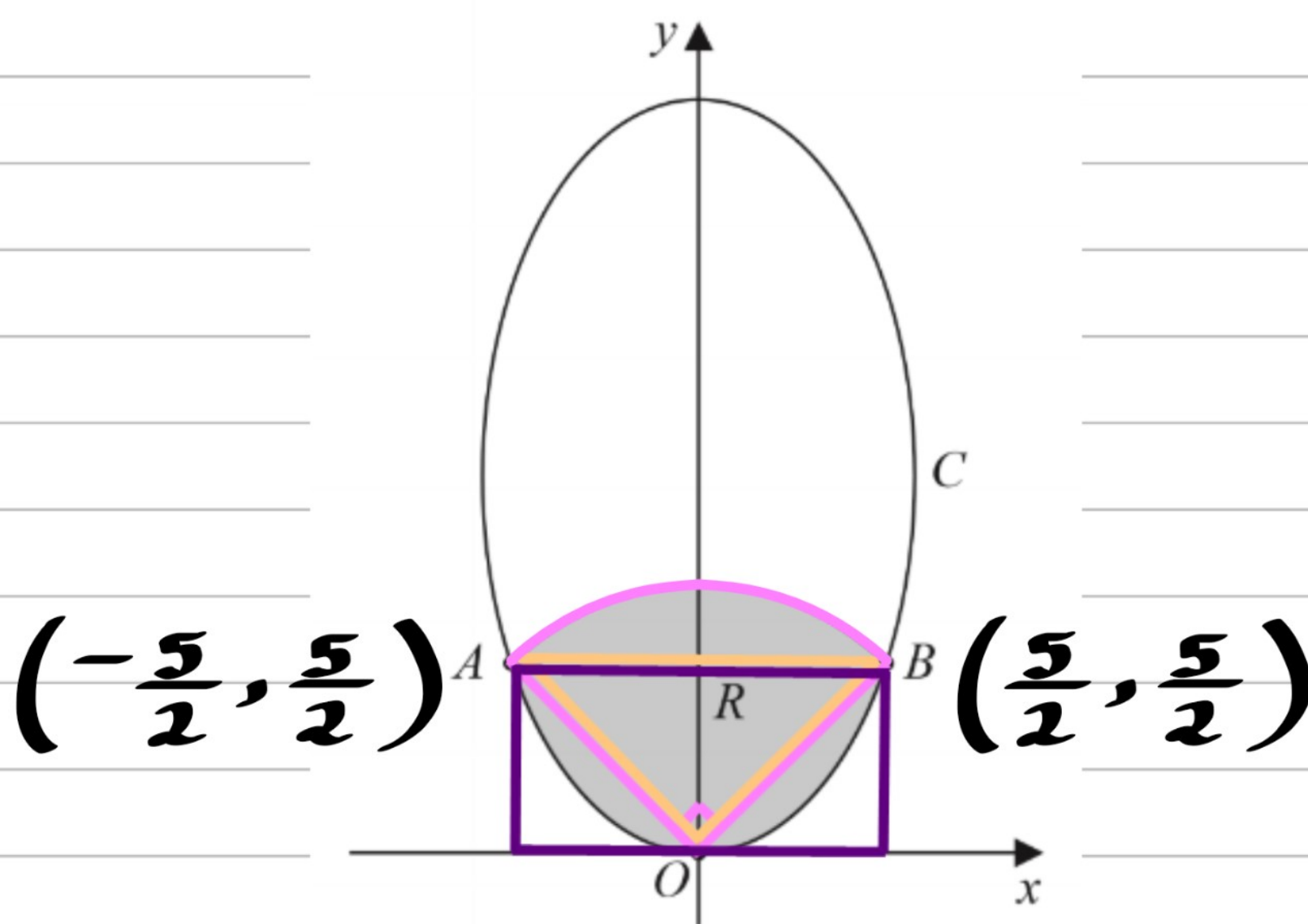
$$= \frac{50}{\sqrt{3}} \left[\frac{3\sqrt{3}}{8} - \frac{\pi}{6} \right]$$

$$= \frac{75}{4} - \frac{25\pi}{3\sqrt{3}}$$

$$t = \frac{\pi}{3} : x = \frac{5}{\sqrt{3}} \sin\left(\frac{\pi}{3}\right) = \frac{5}{2}$$

$$y = 5 \left\{ 1 - \cos\left(\frac{\pi}{3}\right) \right\} = \frac{5}{2}$$

$$\therefore A\left(-\frac{5}{2}, \frac{5}{2}\right) \text{ AND } B\left(\frac{5}{2}, \frac{5}{2}\right)$$



$$\text{AREA OF SECTOR OAB} = \frac{1}{2} \times \frac{\pi}{2} \times \left\{ \frac{5\sqrt{2}}{2} \right\}^2 = \frac{25\pi}{8}$$

$$\text{AREA OF TRIANGLE AOB} = \frac{1}{2} \times \frac{5}{2} \times \frac{5}{2} = \frac{25}{4}$$

$$\text{AREA OF RECTANGLE} = 5 \times \frac{5}{2} = \frac{25}{2}$$



Question 3 continued

$$\text{AREA UNDER CIRCLE} = \frac{25\pi}{8} - \frac{25}{4} + \frac{25}{2}$$

$$= \frac{25\pi}{8} + \frac{25}{4}$$

$$\therefore \text{SHADED AREA} = \frac{25\pi}{8} + \frac{25}{4} - \left\{ \frac{75}{4} - \frac{25\pi}{3\sqrt{3}} \right\}$$

$$= 25\pi \left\{ \frac{1}{8} + \frac{1}{3\sqrt{3}} \right\} - \frac{25}{2}$$

$$= 25 \left[\left(\frac{9 + 8\sqrt{3}}{72} \right) \pi - \frac{1}{2} \right]$$

(Total for Question 3 is 10 marks)



P 7 1 3 8 0 A 0 1 1 3 6

4. (a) Use the trapezium rule with 4 strips to find an approximate value for

$$\int_0^1 16^x \, dx \quad (2)$$

- (b) Use the trapezium rule with n strips to write down an expression that would give an approximate value for

$$\int_0^1 16^x \, dx \quad (2)$$

- (c) Hence show that

$$\int_0^1 16^x \, dx = \lim_{n \rightarrow \infty} \left(\frac{1}{n} \left(1 + 16^{\frac{1}{n}} + \dots + 16^{\frac{n-1}{n}} \right) \right) \quad (3)$$

- (d) Use integration to determine the exact value of

$$\int_0^1 16^x \, dx \quad (3)$$

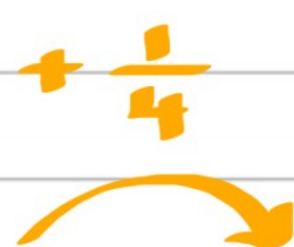
Given that the limit exists,

- (e) use part (c) and the answer to part (d) to determine the exact value of

$$\lim_{x \rightarrow 0} \frac{16^x - 1}{x} \quad (5)$$

(+S1)

(a) LET $y = 16^x$



x	0	$\frac{1}{4}$	$\frac{2}{4}$	$\frac{3}{4}$	1
y	1	2	4	8	16

$$\begin{aligned} \therefore \int_0^1 16^x \, dx &\approx \frac{1}{2} \times \frac{1}{4} \times \{1 + 2(2 + 4 + 8) + 16\} \\ &= 5.625 \end{aligned}$$

Question 4 continued

$+\frac{1}{n}$



x	0	$\frac{1}{n}$	$\frac{2}{n}$	...	$\frac{n-1}{n}$	1
y	1	$16^{\frac{1}{n}}$	$16^{\frac{2}{n}}$	...	$16^{\frac{n-1}{n}}$	16

$$\int_0^1 16^x dx \approx \frac{1}{2} \times \frac{1}{n} \times \left\{ 1 + 2 \left(16^{\frac{1}{n}} + 16^{\frac{2}{n}} + \dots + 16^{\frac{n-1}{n}} \right) + 16 \right\}$$

$$\approx \frac{1}{2n} \left\{ 17 + 2 \left(16^{\frac{1}{n}} + 16^{\frac{2}{n}} + \dots + 16^{\frac{n-1}{n}} \right) \right\}$$

$$(c) \int_0^1 16^x dx = \lim_{n \rightarrow \infty} \frac{1}{2n} \left\{ 17 + 2 \left(16^{\frac{1}{n}} + 16^{\frac{2}{n}} + \dots + 16^{\frac{n-1}{n}} \right) \right\}$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n} \left\{ \frac{17}{2} + 16^{\frac{1}{n}} + 16^{\frac{2}{n}} + \dots + 16^{\frac{n-1}{n}} \right\}$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n} \left\{ \frac{17}{2} - \underbrace{\frac{15}{2}}_0 + \frac{15}{2} + 16^{\frac{1}{n}} + 16^{\frac{2}{n}} + \dots + 16^{\frac{n-1}{n}} \right\}$$

$$= \underbrace{\lim_{n \rightarrow \infty} \frac{1}{n} \left\{ \frac{15}{2} \right\}}_0 + \lim_{n \rightarrow \infty} \frac{1}{n} \left\{ \frac{17}{2} - \frac{15}{2} + 16^{\frac{1}{n}} + 16^{\frac{2}{n}} + \dots + 16^{\frac{n-1}{n}} \right\}$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n} \left\{ 1 + 16^{\frac{1}{n}} + 16^{\frac{2}{n}} + \dots + 16^{\frac{n-1}{n}} \right\}$$

$$(d) \int_0^1 16^x dx = \left[\frac{16^x}{\ln 16} \right]_0^1$$

$$= \frac{15}{\ln 16}$$

Question 4 continued

(e) using (c): $1 + 16^{\frac{1}{n}} + 16^{\frac{2}{n}} + \dots + 16^{\frac{n-1}{n}}$

$$= 1 + 16^{\frac{1}{n}} + \left\{16^{\frac{1}{n}}\right\}^2 + \dots + \left\{16^{\frac{1}{n}}\right\}^{n-1}$$

THIS IS A GEOMETRIC SERIES WITH

$$a = 1, r = 16^{\frac{1}{n}}$$

$$\therefore S_n = 1 + 16^{\frac{1}{n}} + 16^{\frac{2}{n}} + \dots + 16^{\frac{n-1}{n}}$$

$$= \frac{1 \times \left\{(16^{\frac{1}{n}})^n - 1\right\}}{16^{\frac{1}{n}} - 1}$$

$$= \frac{15}{16^{\frac{1}{n}} - 1}$$

$$\therefore \int_0^1 16^x dx = \lim_{n \rightarrow \infty} \frac{1}{n} \left\{ \frac{15}{16^{\frac{1}{n}} - 1} \right\}$$

From (d), $\int_0^1 16^x dx = \frac{15}{\ln 16}$

$$\therefore \frac{15}{\ln 16} = \lim_{n \rightarrow \infty} \frac{1}{n} \left\{ \frac{15}{16^{\frac{1}{n}} - 1} \right\}$$

$$\therefore \frac{15}{\ln 16} = 15 \lim_{n \rightarrow \infty} \left\{ \frac{\frac{1}{n}}{16^{\frac{1}{n}} - 1} \right\}$$

$$\therefore \frac{1}{\ln 16} = \lim_{n \rightarrow \infty} \left\{ \frac{\frac{1}{n}}{16^{\frac{1}{n}} - 1} \right\}$$

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Question 4 continued

LET $x = \frac{1}{n}$, THEN $x \rightarrow 0$ AS $n \rightarrow \infty$

AS LIMIT EXISTS, BY CONTINUITY:

$$\frac{1}{\ln 16} = \lim_{x \rightarrow 0} \left\{ \frac{x}{16^x - 1} \right\}$$

$$\therefore \lim_{x \rightarrow 0} \left\{ \frac{16^x - 1}{x} \right\} = \ln 16$$

(Total for Question 4 is 16 marks)



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5.

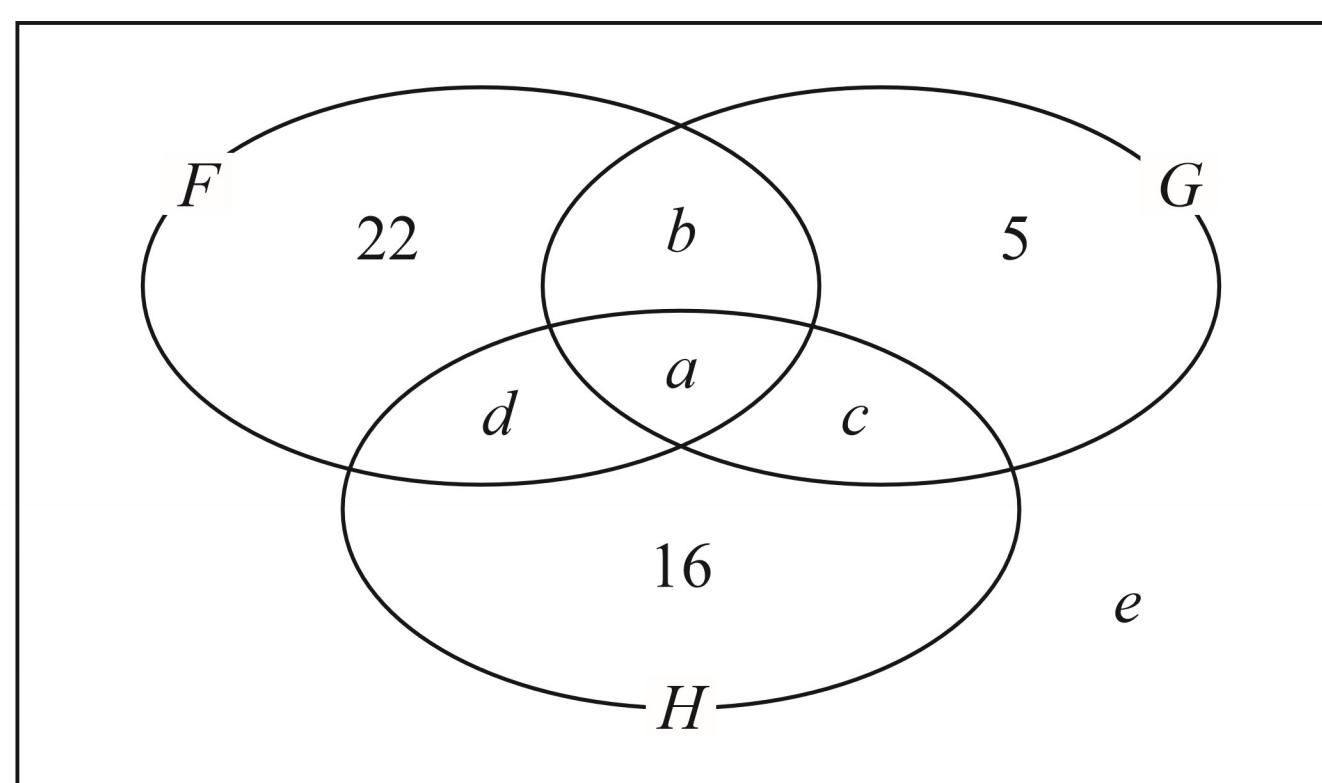


Figure 2

Figure 2 shows a partially completed Venn diagram of sports that a year group of students enjoy, where a , b , c , d and e are non-negative integers.

The diagram shows how many students enjoy a combination of football (F), golf (G) and hockey (H) or none of these sports.

There are n students in the year group.

It is known that

- $P(F) = \frac{3}{7}$
- $P(H | G) = \frac{1}{3}$
- F is independent of $H \cap G$

(a) Show that $P(F \cap H \cap G) = \frac{1}{7} P(G)$ (3)

(b) Prove that if two events X and Y are independent, then X' and Y are also independent. (3)

(c) Hence find the value k such that $P(F' \cap H \cap G) = kP(G)$ (2)

(d) Show that $c = \frac{4}{3}a$ (2)

Given further that $P(F | H) = \frac{1}{5}$

(e) find an expression for d in terms of a , and hence deduce the maximum possible value of a . (3)

(f) Determine the possible values of n . (6)
(+S2)



Question 5 continued

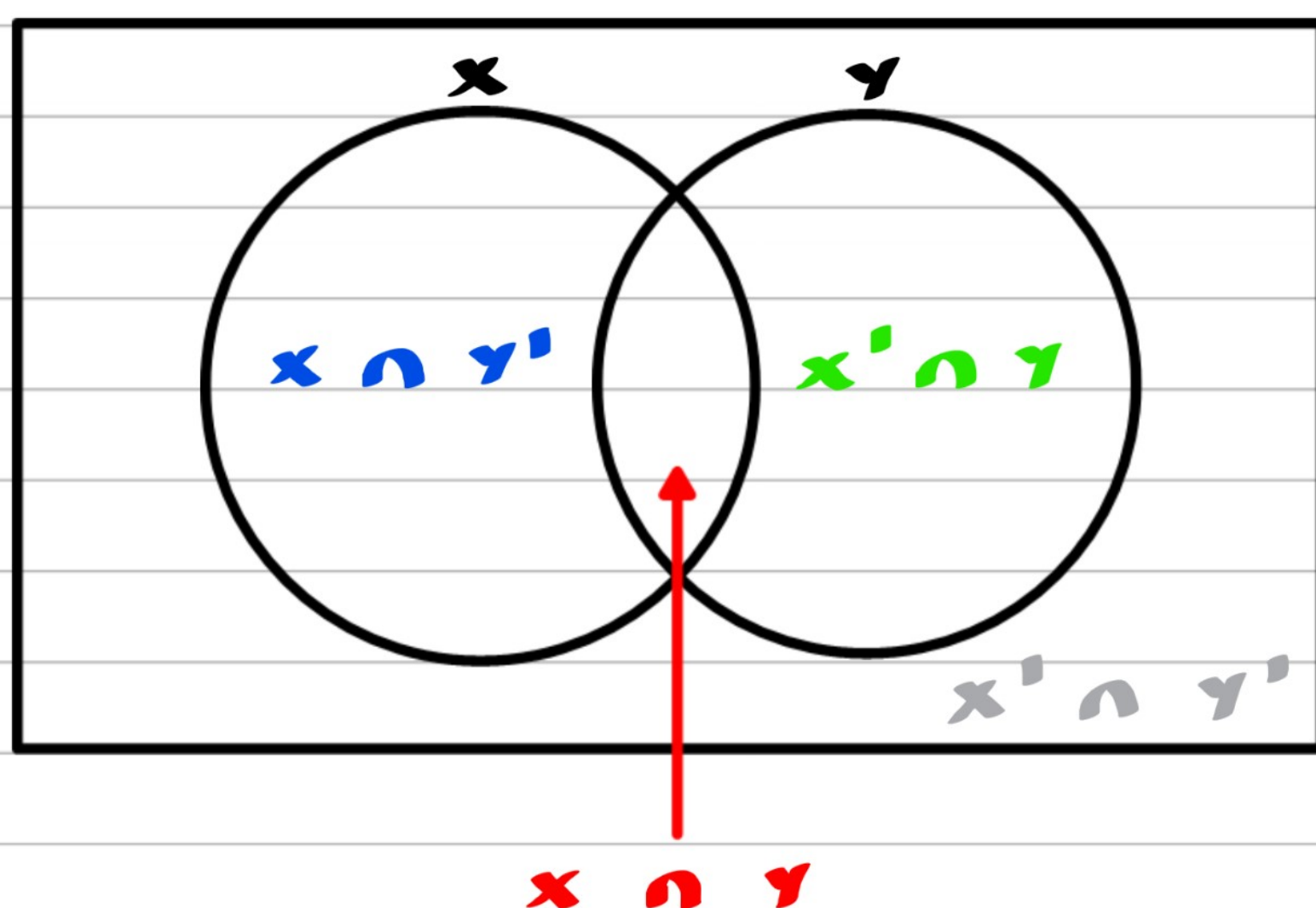
(a)
$$\begin{aligned} P(F \cap G \cap H) &= P(F) \times P(G \cap H) \\ &= P(F) \times P(G) \times P(H|G) \\ &= \frac{3}{7} \times \frac{1}{3} \times P(G) \\ &= \frac{1}{7} P(G) \end{aligned}$$

(b) WE WANT TO SHOW $P(x' \cap y) = P(x')P(y)$

NOTE, $x \cap y$ AND $x' \cap y$ ARE MUTUALLY EXCLUSIVE

IF x AND y ARE INDEPENDENT

$$\therefore P(x \cap y) = P(x)P(y)$$



$$\begin{aligned} P(x' \cap y) &= P(y) - P(x \cap y) \\ &= P(y) - P(x)P(y) \\ &= P(y)\{1 - P(x)\} \\ &= P(x')P(y), \text{ AS REQUIRED} \end{aligned}$$

Question 5 continued

$$\begin{aligned} (c) \quad P(F' \cap G \cap H) &= P(F') P(G \cap H) \\ &= \{1 - P(F)\} \frac{1}{3} P(G) \\ &= \frac{4}{7} \frac{1}{3} P(G) \\ &= \frac{4}{21} P(G) \end{aligned}$$

$$\therefore k = \frac{4}{21}$$

$$(d) \quad P(F' \cap G \cap H) = \frac{c}{n}$$

$$P(G) = \frac{5 + a + b + c}{n}$$

$$\therefore \frac{c}{n} = \frac{4}{21} \left\{ \frac{5 + a + b + c}{n} \right\}$$

$$P(H|G) = \frac{a + c}{5 + a + b + c}$$

$$\therefore \frac{a + c}{5 + a + b + c} = \frac{1}{3}$$

$$\therefore 5 + a + b + c = 3(a + c)$$

$$\therefore \frac{c}{n} = \frac{4}{21} \left\{ \frac{3(a + c)}{n} \right\}$$

$$\therefore 21c = 12(a + c)$$

$$\therefore 7c = 4a + 4c$$

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Question 5 continued

$$\therefore c = \frac{4}{3}a$$

$$(e) P(F|H) = \frac{a + d}{a + c + d + 16}$$

$$\therefore \frac{a + d}{a + c + d + 16} = \frac{1}{5}$$

$$\therefore \frac{a + d}{a + \frac{4}{3}a + d + 16} = \frac{1}{5}$$

$$\therefore 5(a + d) = a + \frac{4}{3}a + d + 16$$

$$\therefore 5a + 5d = \frac{7}{3}a + d + 16$$

$$\therefore 4d = 16 - \frac{8}{3}a$$

$$\therefore d = 4 - \frac{2}{3}a$$

$$\text{But } d \geq 0 : 4 - \frac{2}{3}a \leq 0$$

$$6 \leq a$$

$\therefore$ maximum value of a is 6

$$(f) \text{ we know } c = \frac{4}{3}a$$



Question 5 continued

As $a, b, c, d, e \in \mathbb{Z}^+$, a must be 0, 3, 6

$$P(H|G) = \frac{a+c}{a+b+c+5}$$

$$\therefore \frac{a+c}{a+b+c+5} = \frac{1}{3}$$

$$\therefore 3(a+c) = a+b+c+5$$

$$\therefore b+5 = 2a+2c$$

$$\therefore b+5 = 2a + 2 \times \frac{4}{3}a$$

$$\therefore b = \frac{14}{3}a - 5$$

When $a=0$, $b=-5$, which is impossible

$$a=3 \Rightarrow b=9$$

$$a=6 \Rightarrow b=23$$

$$P(F) = \frac{a+b+d+22}{n}$$

$$= \frac{1}{n} \left\{ a + \frac{14}{3}a - 5 + 4 - \frac{2}{3}a + 22 \right\}$$

$$= \frac{21+5a}{n}$$

$$\therefore \frac{21+5a}{n} = \frac{3}{7}$$

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Question 5 continued

$$\therefore n = \frac{7}{3}(21 + 5a)$$

$$a = 3 \Rightarrow n = 84$$

$$a = 6 \Rightarrow n = 119$$

(Total for Question 5 is 21 marks)



P 7 1 3 8 0 A 0 2 1 3 6

6. In this question you may assume the following formulae for the volume and curved surface area of a cone of base radius r and height h and of a sphere of radius r .
- Cone:** volume $V = \frac{1}{3}\pi r^2 h$ and curved surface area $S = \pi r \sqrt{h^2 + r^2}$
- Sphere:** volume $V = \frac{4\pi}{3}r^3$ and curved surface area $S = 4\pi r^2$

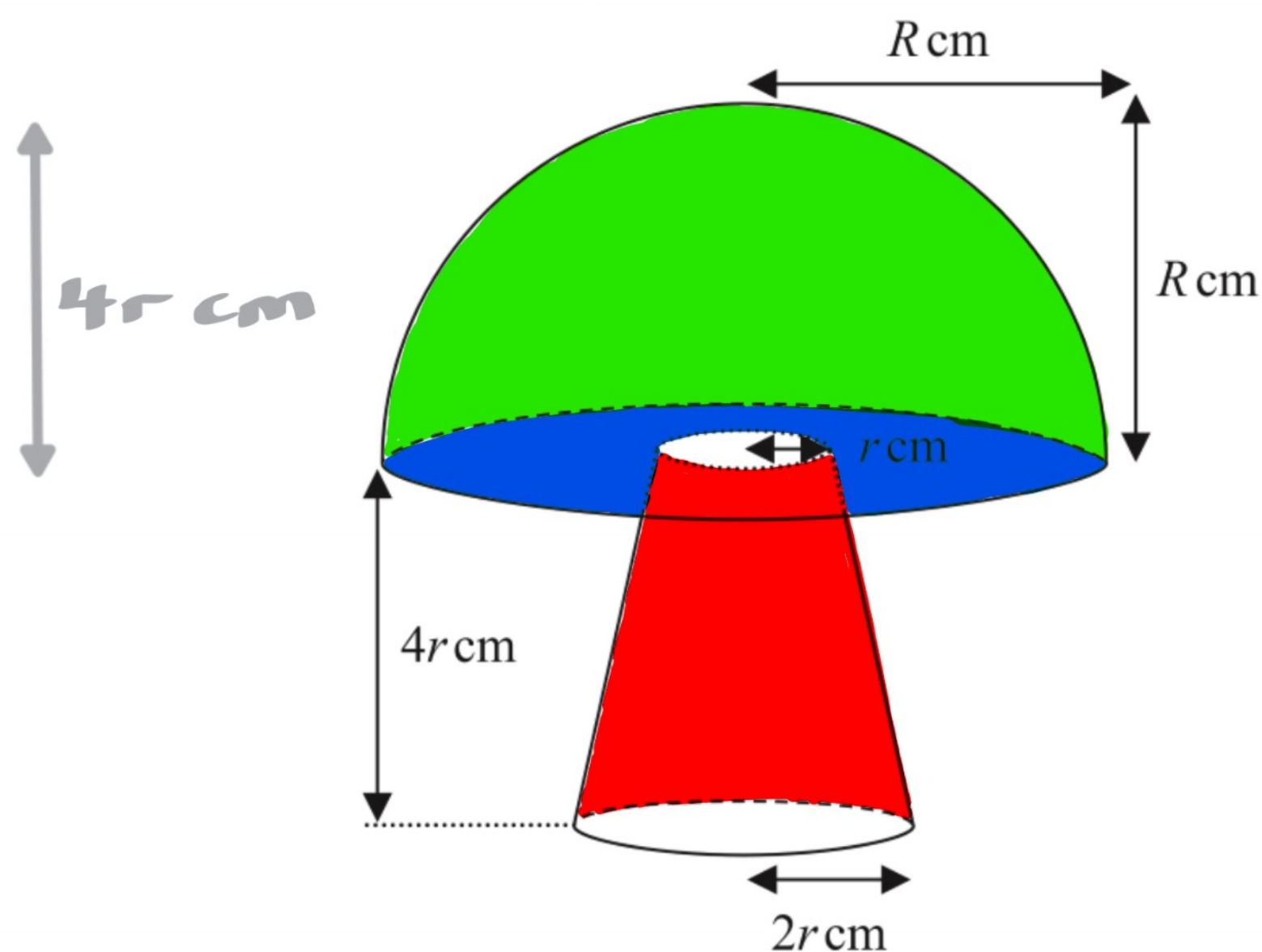


Figure 3

Figure 3 shows the design for a garden ornament.

The ornament is made of a hemisphere on top of a truncated cone.

The truncated cone has base radius $2r$ cm, top radius r cm and height $4r$ cm.

The hemisphere has radius R cm.

Given that the volume of the ornament is 2100π cm³

(a) show that

$$R^3 = 3150 - 14r^3 \quad (5)$$

(b) Find an expression involving $\frac{dR}{dr}$ in terms of r and/or R .

(2)

The base of the truncated cone of the ornament is fixed to the ground.

(c) Show that the visible surface area of the ornament, A cm², is given by

$$A = (3\sqrt{17} - 1)\pi r^2 + 3\pi R^2 \quad (5)$$

Question 6 continued

(d) Hence show that

$$\frac{dA}{dr} = \gamma\pi r - \frac{\delta\pi r^2}{R}$$

where γ and δ are real numbers to be determined.

(2)

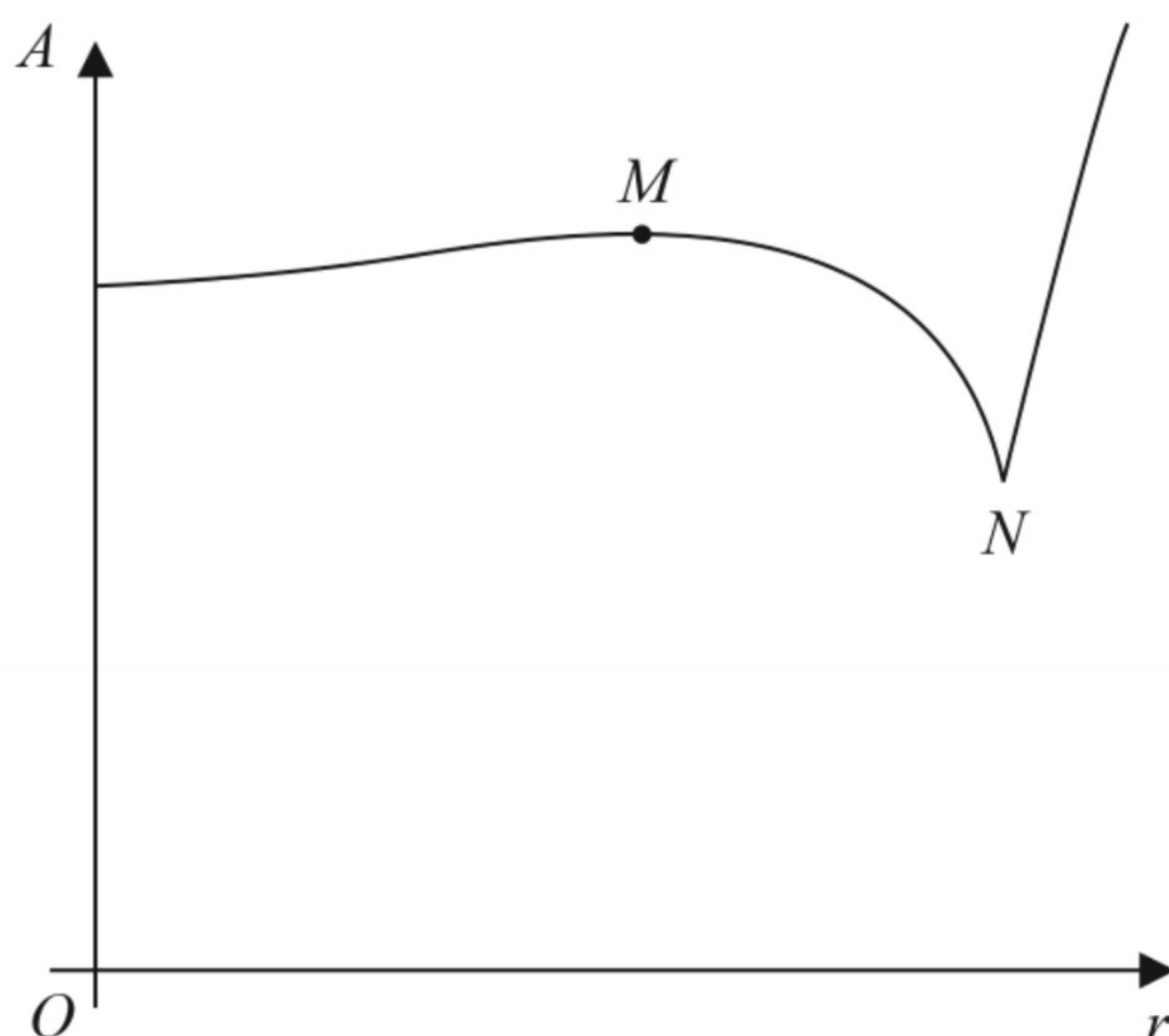


Figure 4

Figure 4 shows a sketch of A against r , for $r \geq 0$

There is a local minimum at $r = 0$ and a local maximum at the point M . The overall minimum point is at the point N , where the gradient of the curve is undefined.

(e) (i) Determine the r coordinate of the point N .

(ii) Explain why, for the ornament, r must be less than this value.

(2)

(f) Show that the r coordinate of the point M is

$$\sqrt[3]{\frac{p(3\sqrt{17}-1)^3}{3q^2 + (3\sqrt{17}-1)^3}}$$

where p and q are integers to be determined.

(5)

(+S2)

$$\text{VOL OF HEMISPHERE} = \frac{1}{2} \times \frac{4}{3} \pi R^3 = \frac{2}{3} \pi R^3$$

VOL OF TRUNCATED CONE

$$= \frac{\pi}{3} \times (2r)^3 \times 8r - \frac{\pi}{3} r^3 \times 4r = \frac{28}{3} \pi r^3$$



Question 6 continued

$$\therefore \frac{28\pi r^3}{3} + \frac{2\pi R^3}{3} = 2100\pi$$

$$\therefore 28r^3 + 2R^3 = 6300$$

$$\therefore R^3 = 3150 - 14r^3$$

(b) $3R^2 \frac{dR}{dr} = -42r^2$

$$\therefore \frac{dR}{dr} = -\frac{14r^2}{R^2}$$

(c) $A = 2\pi R^2 + \pi(R^2 - r^2) +$

$$\pi \times 2r \times \sqrt{(8r)^2 + (2r)^2} - \pi \times r \times \sqrt{(4r)^2 + r^2}$$

$$= 3\pi R^2 - \pi r^2 + 2\pi r \times 2r\sqrt{17} - \pi r \times r\sqrt{17}$$

$$= 3\pi R^2 - \pi r^2 + 4\pi r^2\sqrt{17} - \pi r^2\sqrt{17}$$

$$= (3\sqrt{17} - 1)\pi r^2 + 3\pi R^2$$

(d) $\frac{dA}{dr} = 2(3\sqrt{17} - 1)\pi r + 6\pi R \frac{dR}{dr}$

$$= 2(3\sqrt{17} - 1)\pi r + 6\pi R \times -\frac{14r^2}{R^2}$$

$$= 2(3\sqrt{17} - 1)\pi r - 84\pi \frac{r^2}{R}$$

$$\therefore \gamma = 2(3\sqrt{17} - 1), \delta = -84$$

(e) (i) At ∞ , $R = 0 \therefore 0 = 3150 - 14r^3$

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Question 6 continued

$$r = \sqrt[3]{\frac{3150}{14}}$$

(ii) FOR VALUES OF r GREATER THAN THIS,
 R IS NEGATIVE, WHICH IS NOT POSSIBLE
FOR THE ORNAMENT

(f)

m IS A LOCAL MAXIMUM, i.e.

$$2(3\sqrt{17} - 1)\pi r - 84\pi \frac{r^2}{R} = 0$$

$$\therefore (3\sqrt{17} - 1)R - 42r = 0$$

$$\therefore (3\sqrt{17} - 1)R = 42r$$

$$\therefore R^3 (3\sqrt{17} - 1)^3 = (42r)^3$$

$$\therefore (3150 - 14r^3)(3\sqrt{17} - 1)^3 = 74088r^3$$

$$\therefore 3150(3\sqrt{17} - 1)^3 - 14r^3(3\sqrt{17} - 1)^3 = 74088r^3$$

$$\therefore 3150(3\sqrt{17} - 1)^3 = r^3 \{74088 + 14(3\sqrt{17} - 1)^3\}$$

$$\therefore 225(3\sqrt{17} - 1)^3 = r^3 \{5292 + (3\sqrt{17} - 1)^3\}$$

$$r^3 = \frac{225(3\sqrt{17} - 1)^3}{5292 + (3\sqrt{17} - 1)^3}$$

$$r^3 = \frac{225(3\sqrt{17} - 1)^3}{3 \times 42^2 + (3\sqrt{17} - 1)^3}$$

$$\therefore r = \sqrt[3]{\frac{225(3\sqrt{17} - 1)^3}{3 \times 42^2 + (3\sqrt{17} - 1)^3}}$$

$$\therefore p = 225, q = 42$$

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7. A sequence of non-zero real numbers $a_1, a_2, a_3, \dots$ is defined by

$$a_{n+1} = p + \frac{q}{a_n} \quad n \in \mathbb{N}$$

where p and q are real numbers with $q \neq 0$

It is known that

- one of the terms of this sequence is a
- the sequence is periodic

(a) Determine an equation for q , in terms of p and a , such that the sequence is constant (of period/order one).

(2)

(b) Determine the value of p that is necessary for the sequence to be of period/order 2.

(3)

(c) Give an example of a sequence that satisfies the condition in part (b), but is **not** of period/order 2.

(2)

(d) Determine an equation for q , in terms of p only, such that the sequence has period/order 4.

(7)

(+S1)

(a) $a_{n+1} = p + \frac{q}{a_n}, q \neq 0$

REQUIRE $a_{n+1} = a_n = a$

$\therefore p + \frac{q}{a} = a$

$\therefore q = a^2 - ap$

(b) a_1

$a_2 = p + \frac{q}{a_1}$

$a_3 = p + \frac{q}{a_2} = p + \frac{q}{p + \frac{q}{a_1}}$



Question 7 continued

PERIOD OF 2 $\therefore a_1 = a_3$

$$\therefore p + \frac{q}{p + \frac{q}{a_1}} = a_1$$

$$\therefore p \left\{ p + \frac{q}{a_1} \right\} + q = a_1 \left\{ p + \frac{q}{a_1} \right\}$$

$$\therefore a_1 p^2 + pq + a_1 q = a_1^2 p + a_1 q$$

$$\therefore a_1 p^2 + pq - a_1^2 p = 0$$

$$\therefore p(a_1 p + q - a_1^2) = 0$$

$$\therefore \text{IF } p = \frac{a_1^2 - q}{a_1}, \text{ THEN SEQUENCE IS CONSTANT}$$

$$\therefore p = 0$$

(c) REQUIRE SEQUENCE OF ORDER 1

$$\therefore a_{n+1} = \frac{q}{a_n}$$

$$a_1 = a, \quad q = a^2$$

$\therefore$ SEQUENCE IS CONSTANT: $a, a, a, \dots$

(d) a_1

$$a_2 = p + \frac{q}{a_1}$$



Question 7 continued

$$a_3 = p + \frac{q}{a_2} = p + \frac{q}{p + \frac{q}{a_1}}$$

$$a_4 = p + \frac{q}{a_3} = p + \frac{q}{p + \frac{q}{p + \frac{q}{a_1}}}$$

$$a_5 = p + \frac{q}{a_4} = p + \frac{q}{p + \frac{q}{p + \frac{q}{p + \frac{q}{a_1}}}}$$

PERIOD OF 4 $\therefore a_1 = a_5$

$$\therefore p + \frac{q}{p + \frac{q}{p + \frac{q}{p + \frac{q}{a_1}}}} = a_1$$

$$\therefore p \left\{ p + \frac{q}{p + \frac{q}{p + \frac{q}{a_1}}} \right\} + q = a_1 \left\{ p + \frac{q}{p + \frac{q}{a_1}} \right\}$$

$$\therefore p \left\{ p \left\{ p + \frac{q}{p + \frac{q}{a_1}} \right\} + q \right\} + q \left\{ p + \frac{q}{a_1} \right\}$$

$$= a_1 \left\{ p \left\{ p + \frac{q}{a_1} \right\} + q \right\}$$

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Question 7 continued

$$\therefore p \{ p \{ p \{ p + \frac{q}{a_1} \} + q \} + q \{ p + \frac{q}{a_1} \} \}$$

$$+ q \{ p \{ p + \frac{q}{a_1} \} + q \}$$

$$= a_1 \{ p \{ p \{ p + \frac{q}{a_1} \} + q \} + q \{ p + \frac{q}{a_1} \} \}$$

$$\therefore p \{ p \{ p \{ a_1 p + q \} + q \{ a_1 p + q \} + q \{ p \{ a_1 p + q \} + a_1 q \} \}$$

$$= a_1 \{ p \{ p \{ a_1 p + q \} + a_1 q \} + q \{ a_1 p + q \} \}$$

$$\therefore a_1 p^4 + p^3 q + 3 a_1 p^2 q + 2 q^2 p + a_1 q^2 = a_1^2 p^3$$

$$+ 2 a_1 p^2 q + a_1^2 q p + a_1 q^2$$

$$\therefore a_1 p^4 + p^3 q + 2 a_1 p^2 q + 2 q^2 p - a_1^2 p^3 - 2 a_1^2 q p = 0$$

$$\therefore p \{ a_1 p^3 + p^2 q - p^2 a_1 + 2 a_1 p q + 2 q^2 - 2 a_1^2 q \} = 0$$

$$\therefore p \{ a_1 p + q - a_1^2 \} \{ p^2 + 2 q \} = 0$$

$$\therefore p = 0 \quad \text{ORDER 2}$$

$$\therefore a_1 p + q - a_1^2 = 0 \quad \text{ORDER 1}$$

$$\therefore q = -\frac{p^2}{2} \quad \text{ORDER 4}$$

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